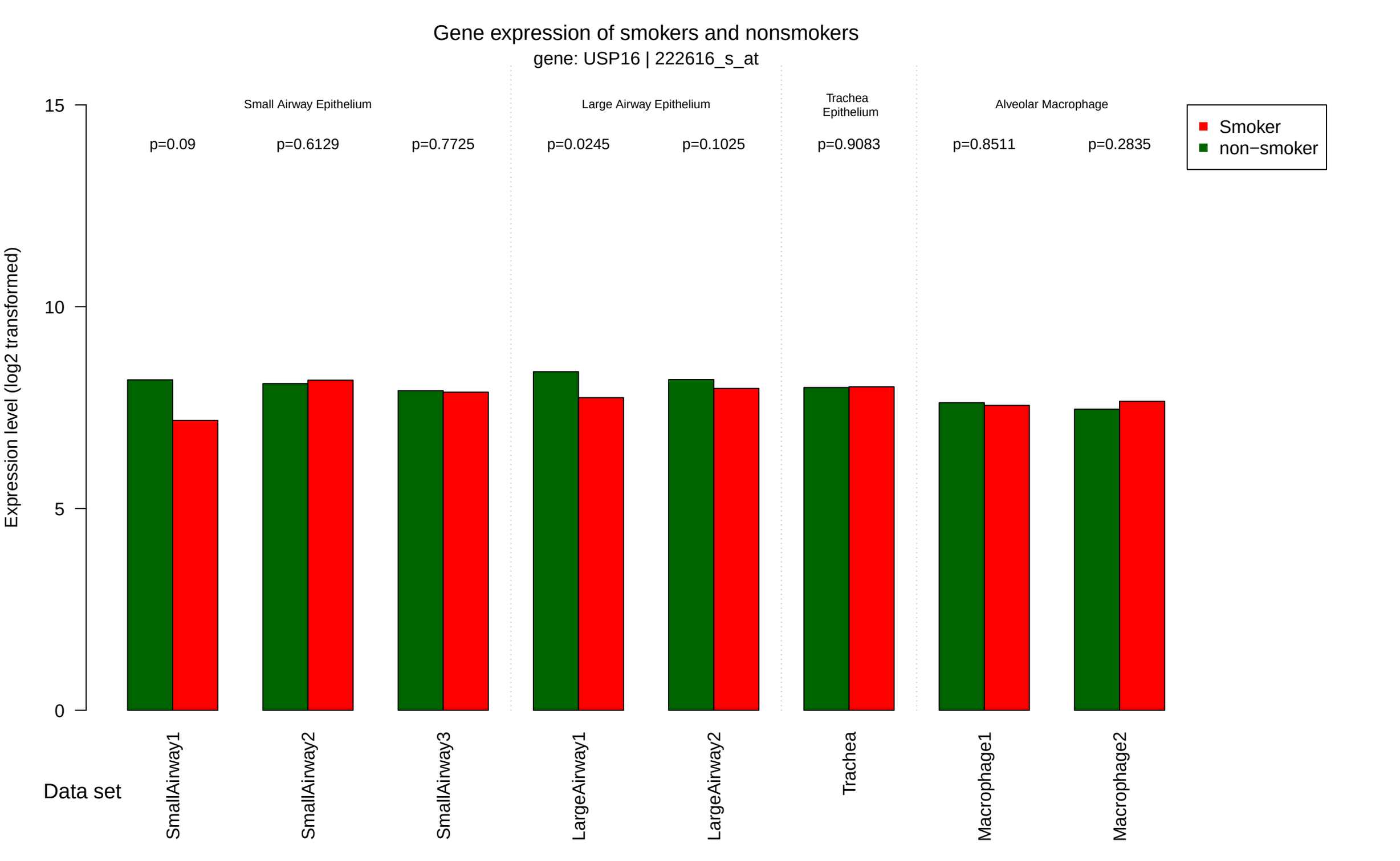
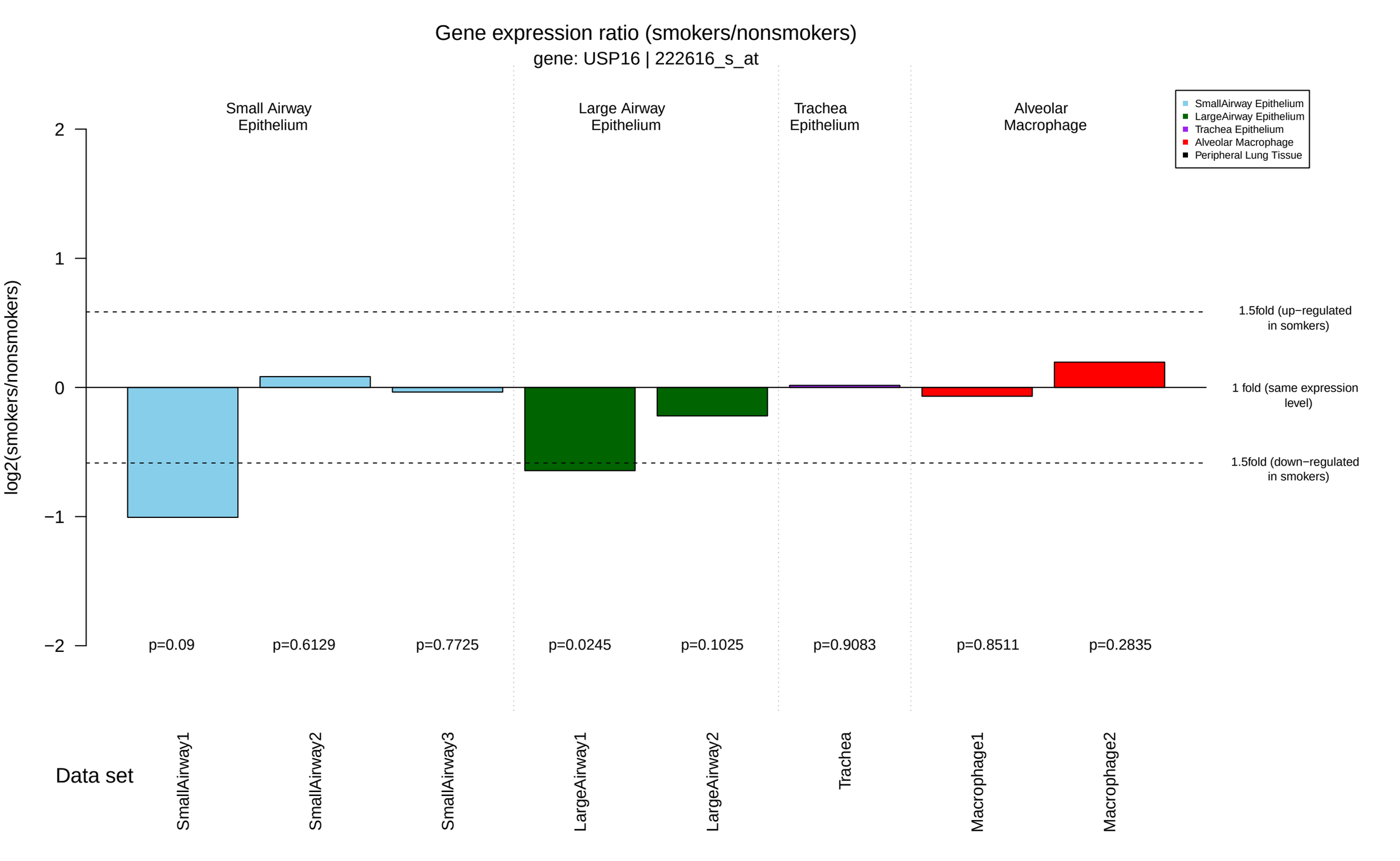
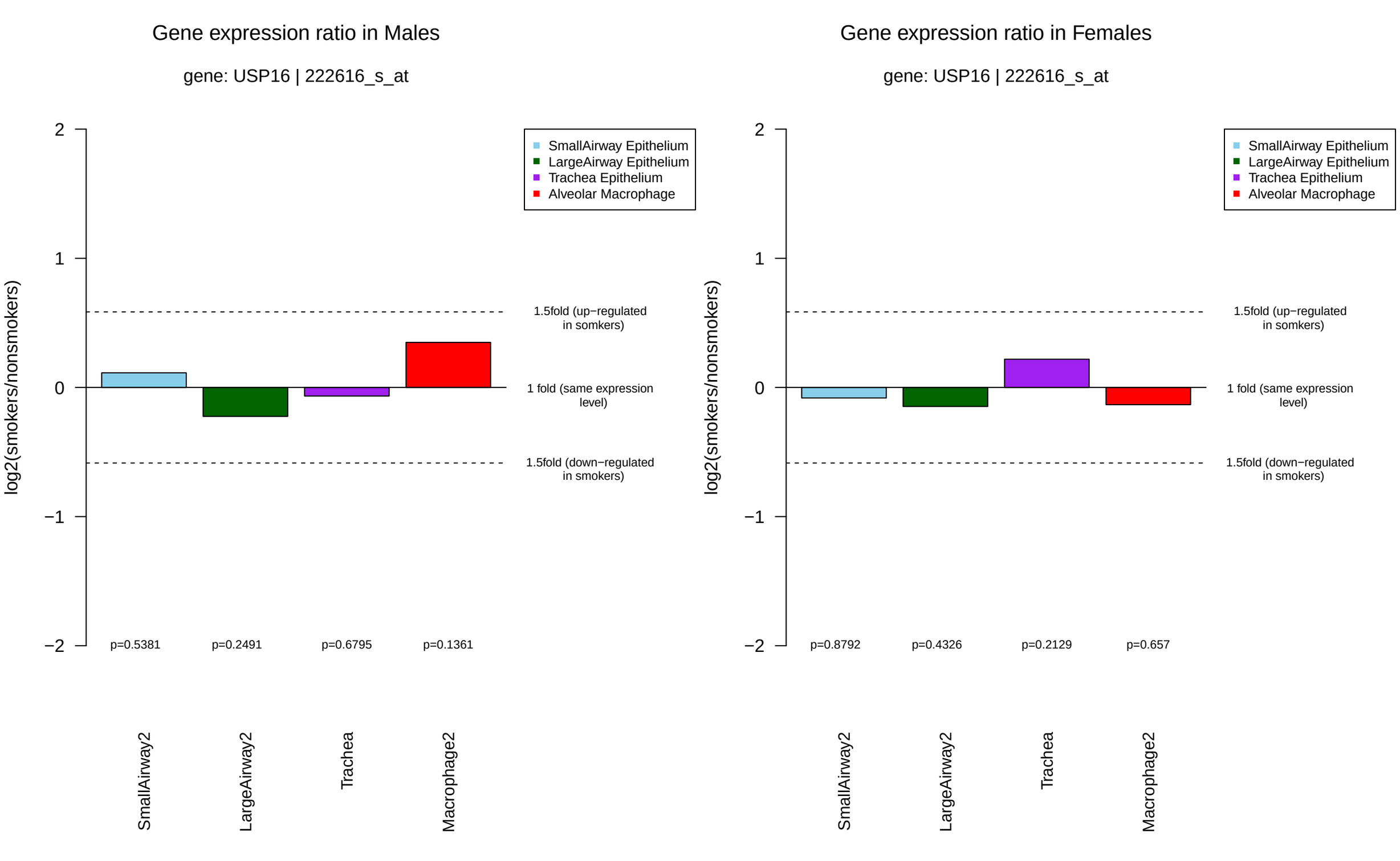
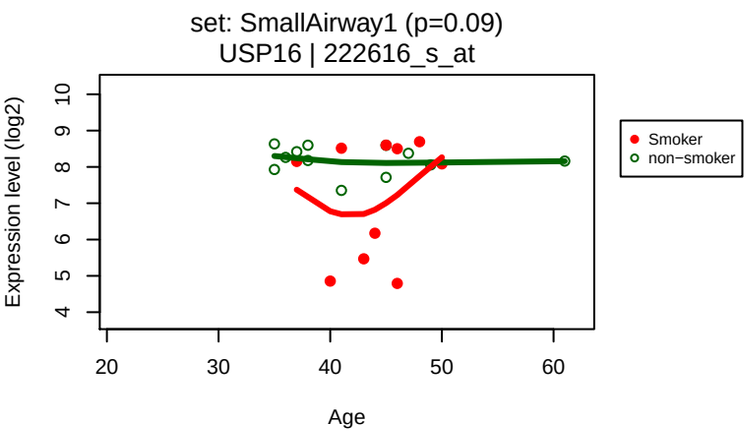
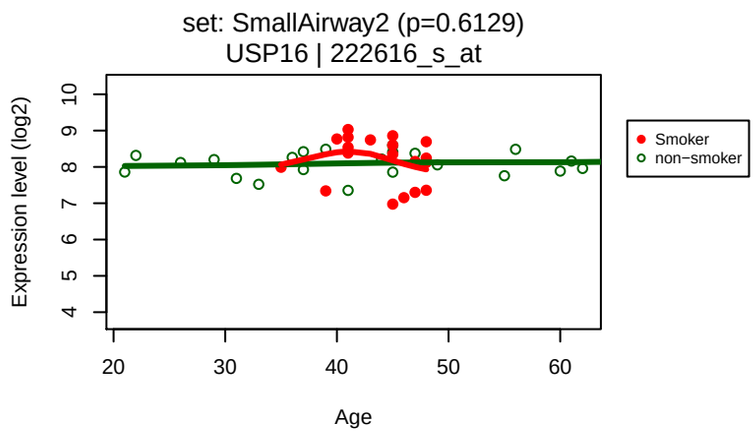
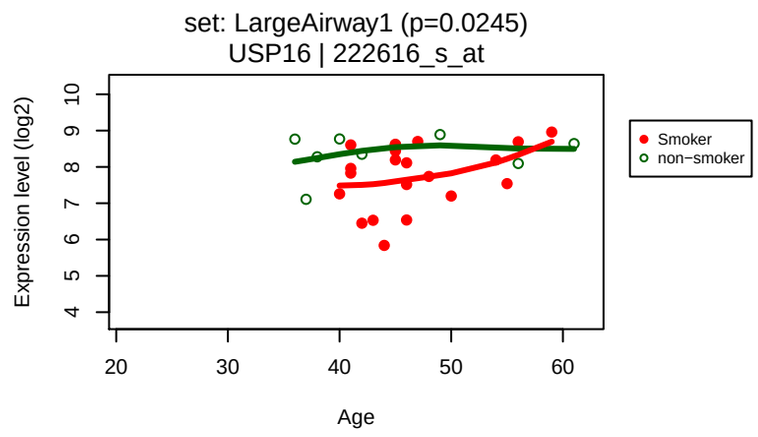

Gene expression of smokers and nonsmokers
gene: USP16 | 222616_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.09
p=0.6129
p=0.7725
p=0.0245
p=0.1025
p=0.9083
p=0.8511
p=0.2835
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

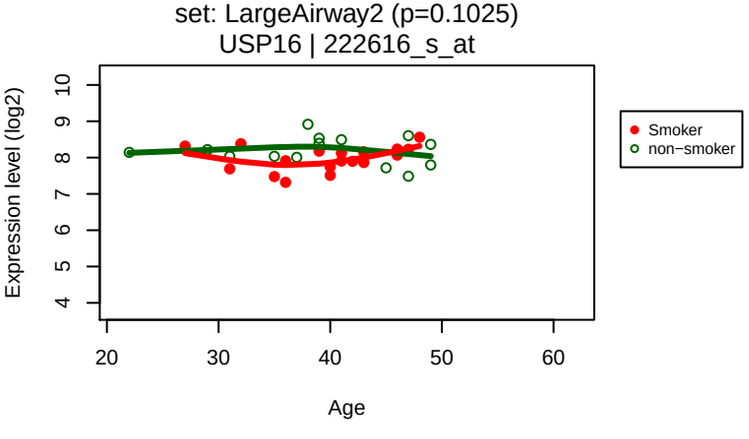

Gene expression ratio (smokers/nonsmokers)
gene: USP16 | 222616_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.09
p=0.6129
p=0.7725
p=0.0245
p=0.1025
p=0.9083
p=0.8511
p=0.2835
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

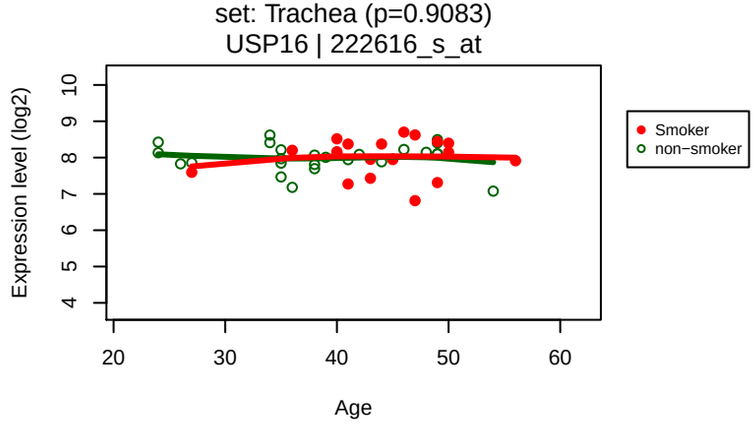

Gene expression ratio in Males
gene: USP16 | 222616_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5381
p=0.2491
p=0.6795
p=0.1361
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP16 | 222616_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8792
p=0.4326
p=0.2129
p=0.657
SmallAirway2
LargeAirway2
Trachea
Macrophage2

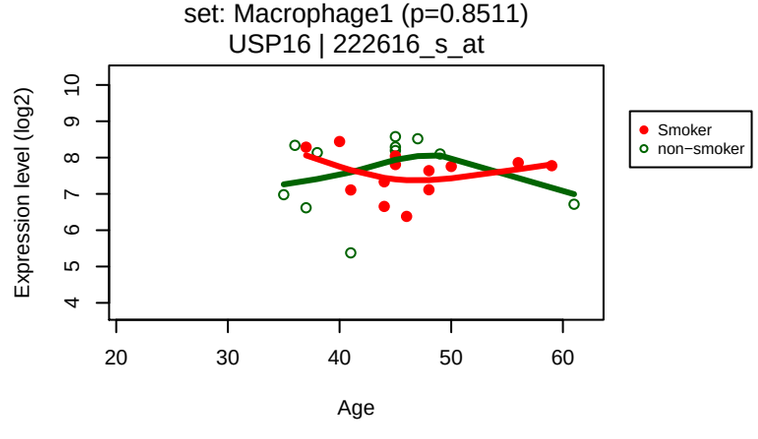

set: SmallAirway1 (p=0.09)
USP16 | 222616_s_at
Expression level (log2)
Age
Smoker
non−smoker

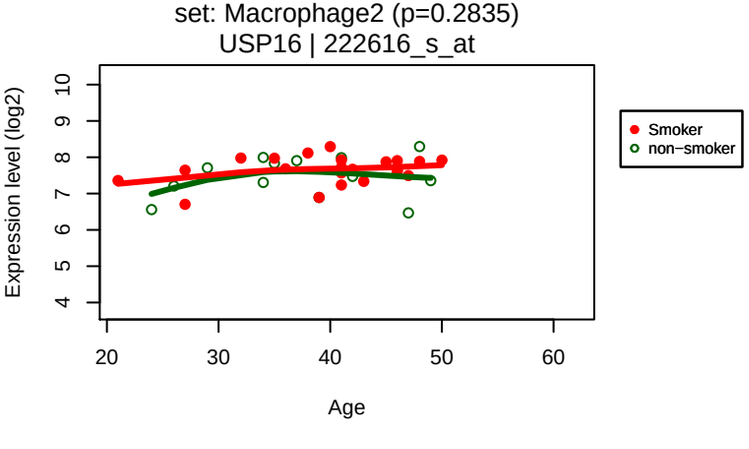

set: SmallAirway2 (p=0.6129)
USP16 | 222616_s_at
Expression level (log2)
Age
Smoker
non−smoker

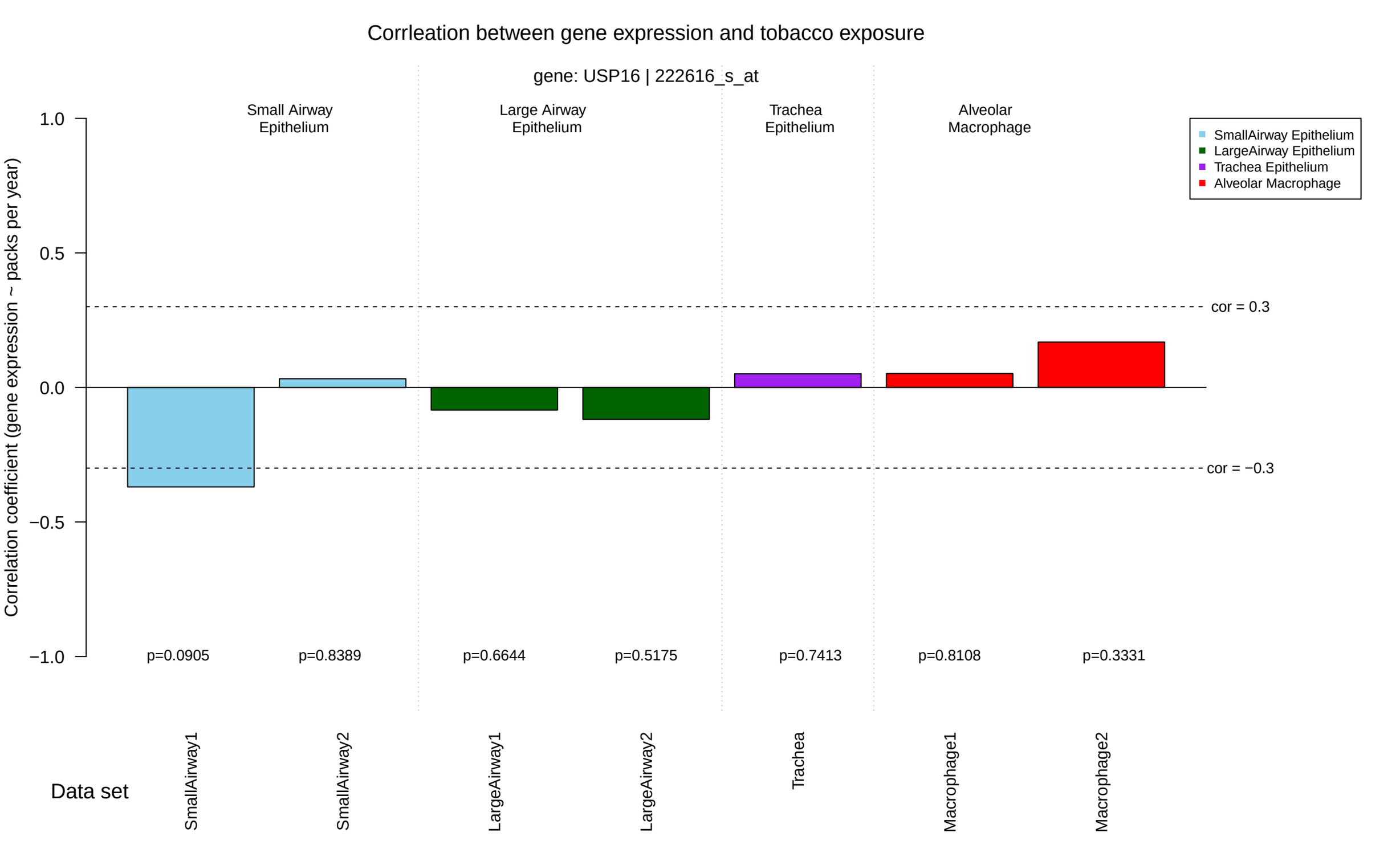

set: LargeAirway1 (p=0.0245)
USP16 | 222616_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1025)
USP16 | 222616_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9083)
USP16 | 222616_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8511)
USP16 | 222616_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2835)
USP16 | 222616_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP16 | 222616_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0905
p=0.8389
p=0.6644
p=0.5175
p=0.7413
p=0.8108
p=0.3331
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2